Deterministic gaps between positive factorization cost and the entrywise norm

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Abstract

For a positive-semidefinite matrix, the sum of the squared ℓ_1 norms of its positive rank-one factors is at least the entrywise absolute sum of the matrix. We construct rational orthogonal projectors for which the minimum factorization cost exceeds this lower bound by a growing factor. Applying the deterministic almost Euclidean subspaces of Guruswami, Lee and Razborov gives an algorithm using polynomially many bit operations in the matrix dimension n, with ratio at least $\sqrt{n}\exp[-O(\log\log n\ \log\log\log n)]$. The estimate holds for both real and complex factorizations. The proof uses the fact that positivity confines every factor to the projector range, together with a trace estimate; complexification preserves the subspace inequality without loss. An elementary construction from additive characters of the cubic graph over $\mathbb{F}_q$, where $q = 2^s$, gives rank-q projectors in dimension q^2 with ratio at least $q^{1/2}/3$. These elementary projectors also specify a rational positive operator with absolutely summable entries for which every countable positive rank-one factorization has infinite total squared ℓ_1 cost.

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1 The factorization question

Let A be a positive-semidefinite matrix over $\mathbb{F}$, where $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$. Following the factorization problem studied by Balan and Jiang [1], define

$$\gamma_+^{\mathbb{F}}(A) = \inf\left\{\sum_{k=1}^{m} \|x_k\|_1^2 : A = \sum_{k=1}^{m} x_k x_k^*, \quad x_k \in \mathbb{F}^n, \quad m \geq 1\right\},$$
$$\|A\|_{1,1} = \sum_{i,j=1}^{n} |A_{ij}|. \tag{1}$$

Here $*$ denotes transpose or conjugate transpose, as appropriate. The factors have arbitrary signs or phases, and their number is unrestricted. Positivity refers to each matrix $x_k x_k^*$; it does not require nonnegative coordinates. Since $\|xx^*\|_{1,1} = \|x\|_1^2$, the triangle inequality gives

$$\|A\|_{1,1} \leq \gamma_+^{\mathbb{F}}(A). \tag{2}$$

The size of the reverse comparison measures how much cancellation between the entries of different positive rank-one terms can reduce the norm of their sum. A lower bound on $\gamma_+^{\mathbb{F}}(A)$ must therefore apply to every decomposition, rather than to a selected eigenbasis or another particular choice of factors.

Bandeira, Mixon and Steinerberger proved a probabilistic lower bound of order $\sqrt{n}$ for this ratio [2]. Their proof includes a primal matrix of the form $aI + bW$, with W a random symmetric sign matrix, and uses probabilistic estimates to obtain suitable signs. Balan and Jiang subsequently used finite-dimensional witnesses in a block construction to obtain a positive operator with absolutely summable entries and no factorization of finite total squared ℓ_1 cost [1, Theorem 5.5]. Their Remark 5.6 asks for explicit finite witnesses. The constructive question considered here is whether a deterministic algorithm can produce rational matrices with a quantitatively large ratio in time polynomial in their dimension.

We use subspaces whose vectors have a uniformly large ℓ_1 norm relative to their Euclidean norm. Orthogonal projection retains this coordinate-dependent property: positivity forces every factor of the projector into the subspace, where its cost can be bounded. The pertinent comparison between nullspace projectors, rank-one matrices, trace and entrywise norm already appears in d'Aspremont's work on sparse-recovery relaxations [3, Section 4.1]. Our use of that geometry is an application to the positive factorization problem. The quantitative construction comes from the published almost Euclidean subspaces of Guruswami, Lee and Razborov [4].

Theorem 1 (deterministic rational witnesses). There is a deterministic algorithm which, for every integer $n \geq 1$, outputs a nonzero rational orthogonal projector $P_n \in \mathbb{Q}^{n\times n}$ using a number of bit operations polynomial in n. There are constants $C > 0$ and n_0 such that, for $n \geq n_0$, its rank is at least $n/2$ and

$$\frac{\gamma_+^{\mathbb{F}}(P_n)}{\|P_n\|_{1,1}} \geq \sqrt{n}\exp[-C\log\log n\ \log\log\log n] = n^{1/2-o(1)}, \qquad \mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}. \tag{3}$$

The running time is polynomial in n, the dimension of the output matrix, rather than in the length of the binary encoding of n. The guarantee supplies deterministic witnesses with a subpolynomial loss from the probabilistic square-root bound. It neither gives a deterministic constant multiple of $\sqrt{n}$ nor determines the largest possible ratio in dimension n.

We first prove the projector estimate and use it on an elementary character construction. This gives a complete finite family with a weaker power-law gap and shows directly how the subspace property controls all decompositions. We then obtain Theorem 1 from [4] and exact rational linear algebra. The elementary family alone suffices for the fully specified infinite operator in Section 5.

2 Subspace inequalities and positive factors

For a real subspace $E \subseteq \mathbb{R}^n$, write $E_{\mathbb{C}} = E + iE$ for its complexification. We measure its departure from the normalized ℓ_1–ℓ_2 inequality by a number $D \geq 1$ satisfying

$$\|x\|_1 \geq \frac{\sqrt{n}}{D}\|x\|_2 \qquad (x \in E). \tag{4}$$

For fixed D, (4) requires every Euclidean unit vector in E to have ℓ_1 norm of order $\sqrt{n}$. The dimension of E will also matter: the factorization lower bound accumulates the squared Euclidean norms of all factors, and their sum is the rank of the projector.

Lemma 2 (projector estimate). Suppose that $E \subseteq \mathbb{R}^n$ has dimension $r \geq 1$ and satisfies (4). If P is the orthogonal projector onto E, then

$$\gamma_+^{\mathbb{F}}(P) \geq \frac{nr}{D^2}, \qquad \|P\|_{1,1} \leq n\sqrt{r}, \qquad \frac{\gamma_+^{\mathbb{F}}(P)}{\|P\|_{1,1}} \geq \frac{\sqrt{r}}{D^2} \tag{5}$$

for both $\mathbb{F} = \mathbb{R}$ and $\mathbb{F} = \mathbb{C}$.

Proof. The same subspace inequality holds on $E_{\mathbb{C}}$ with the same constant. Indeed, if $z = u + iv$ with $u, v \in E$, the triangle inequality in $\mathbb{R}^2$, applied to the vectors $(|u_i|, |v_i|)$, gives

$$\begin{aligned} \|z\|_1 &= \sum_{i=1}^{n} \sqrt{u_i^2 + v_i^2} \\ &\geq \sqrt{\|u\|_1^2 + \|v\|_1^2} \geq \frac{\sqrt{n}}{D}\sqrt{\|u\|_2^2 + \|v\|_2^2} = \frac{\sqrt{n}}{D}\|z\|_2. \end{aligned} \tag{6}$$

Now take any decomposition $P = \sum_k x_k x_k^*$ over either field. For $y \in \ker P$,

$$0 = y^* P y = \sum_k |y^* x_k|^2. \tag{7}$$

Each summand is nonnegative, so each factor is orthogonal to $\ker P$. Thus $x_k \in E$ in the real case and $x_k \in E_{\mathbb{C}}$ in the complex case. Applying (4) or (6), and then taking the trace, yields

$$\sum_k \|x_k\|_1^2 \geq \frac{n}{D^2}\sum_k \|x_k\|_2^2 = \frac{n}{D^2}\operatorname{tr} P = \frac{nr}{D^2}. \tag{8}$$

On the other hand, Cauchy–Schwarz applied to the n^2 entries gives

$$\|P\|_{1,1} \le n\|P\|_F = n\sqrt{\operatorname{tr}(P^*P)} = n\sqrt{r}. \tag{9}$$

Taking the infimum in (8) and using (9) proves (5). □

The argument separates the two quantities in (2) without finding an optimal decomposition. Its lower bound uses the range and trace of the matrix; its upper bound uses the Frobenius norm of a projector. More generally, the same proof gives $\gamma_+^{\mathbb{F}}(X) \ge (n/D^2)\operatorname{tr} X$ for every positive-semidefinite X whose range lies in E or $E_{\mathbb{C}}$. This is the factorization form of the section geometry used in [3]. The lossless passage in (6) concerns the lower bound; no equality between the real and complex optimization values is needed.

We shall also use (8) for countably many factors. If $aP = \sum_{k\ge1} x_k x_k^*$ for $a \ge 0$, with convergence of quadratic forms in the finite-dimensional space, (7) still confines every factor to the range when $a > 0$. Summing the diagonal entries gives $\sum_k \|x_k\|_2^2 = ar$, and consequently

$$\sum_{k\ge1} \|x_k\|_1^2 \ge \frac{anr}{D^2}. \tag{10}$$

For $a = 0$ all factors vanish. Nonnegativity justifies the sums, including the possibility of infinite cost.

3 An elementary family from cubic characters

The estimate in (5) gives a growing ratio whenever the rank grows while the distortion remains bounded. Additive characters provide such a family in which both the projector and its norm estimate can be written explicitly. Let $q = 2^s$ with $s \ge 1$, let $K = \mathbb{F}_q$, and write $\mathrm{Tr} : K \to \mathbb{F}_2$ for the absolute field trace. Index the q^2 coordinates by $g = (a, b) \in K^2$ and define q real characters by

$$\chi_t(a,b) = (-1)^{\mathrm{Tr}(at+bt^3)}, \qquad U_{g,t} = q^{-1}\chi_t(g), \qquad t \in K. \tag{11}$$

The columns have norm one because each contains q^2 entries of absolute value q^{-1}. Distinct columns are orthogonal: if $t \ne u$, the coefficient $t + u$ of a is nonzero, and summing $(-1)^{\mathrm{Tr}(a(t+u)+b(t^3+u^3))}$ over a gives zero. Hence $U^{\mathsf{T}}U = I_q$, and

$$P_q = UU^{\mathsf{T}}, \qquad (P_q)_{g,h} = q^{-2}\sum_{t\in K} \chi_t(g)\chi_t(h) \tag{12}$$

is a rational orthogonal projector of rank q.

The cubic graph $\{(t, t^3) : t \in K\}$ is a classical Sidon set in the additive group K^2 [5, Section 1]. The property needed for the norm estimate is its restriction on four-term additive relations, which we verify directly. Suppose

$$t + u = v + w = d, \qquad t^3 + u^3 = v^3 + w^3. \tag{13}$$

If $d = 0$, then $t = u$ and $v = w$. If $d \ne 0$, the identity $t^3 + u^3 = d^3 + tud$ in characteristic two implies $tu = vw$. Both unordered pairs are therefore the roots of $X^2 + dX + tu$, and $\{t, u\} = \{v, w\}$. In particular, four distinct points of the cubic

graph cannot sum to zero. Every zero-sum quadruple can be partitioned into two pairs of equal points.

To see what this relation gives for the range of U, take complex coefficients c_t, let g be uniform on K^2, and set

$$Y(g) = \sum_{t \in K} c_t \chi_t(g), \qquad \sigma^2 = \sum_{t \in K} |c_t|^2. \tag{14}$$

Character orthogonality gives $\mathbb{E}|Y|^2 = \sigma^2$. In the fourth moment, a term survives exactly when its four graph points sum to zero. The three pairings contribute σ^4, σ^4 and $|\sum_t c_t^2|^2$. Terms with all four indices equal have been counted three times, so subtracting two copies gives

$$\mathbb{E}|Y|^4 = 2\sigma^4 + \left|\sum_t c_t^2\right|^2 - 2\sum_t |c_t|^4 \le 3\sigma^4. \tag{15}$$

Hölder's inequality applied to $|Y|^{2/3}|Y|^{4/3}$ now yields

$$\sigma^2 = \mathbb{E}|Y|^2 \le (\mathbb{E}|Y|)^{2/3}(\mathbb{E}|Y|^4)^{1/3}, \qquad \mathbb{E}|Y| \ge \frac{\sigma}{\sqrt{3}}. \tag{16}$$

The last inequality is immediate when $\sigma = 0$; otherwise it follows by rearranging the preceding bound. For $z = Uc$, orthonormality gives $\|z\|_2 = \sigma$, while the normalization of the uniform measure gives $\|z\|_1 = q\mathbb{E}|Y|$. Consequently

$$\|z\|_1 \ge \frac{q}{\sqrt{3}}\|z\|_2 \qquad (z \in \operatorname{range}_{\mathbb{C}} U). \tag{17}$$

Proposition 3 (elementary rational witnesses). For $q = 2^s$, the projector in (12) has ambient dimension $n = q^2$, rank q, and satisfies

$$\gamma_+^{\mathbb{F}}(P_q) \ge \frac{q^3}{3}, \qquad \|P_q\|_{1,1} \le q^{5/2}, \qquad \frac{\gamma_+^{\mathbb{F}}(P_q)}{\|P_q\|_{1,1}} \ge \frac{\sqrt{q}}{3} = \frac{n^{1/4}}{3} \tag{18}$$

for both fields. Its entries can be computed exactly by a deterministic algorithm in time polynomial in n.

Proof. Equation (17) is the hypothesis of Lemma 2 with $D = \sqrt{3}$, $n = q^2$ and $r = q$, proving all three inequalities. For the algorithm, fix the field representation as follows. Order monic degree-s polynomials $X^s + \sum_{i=0}^{s-1} a_i X^i$ by the integer $\sum_{i=0}^{s-1} a_i 2^i$, choose the first irreducible one, and represent K by the quotient of $\mathbb{F}_2[X]$ by that polynomial. Enumerate its elements by the same binary order. Exhaustive testing for divisors of degree at most $\lfloor s/2 \rfloor$ supplies a deterministic irreducibility test; the entire search takes time polynomial in q. Evaluating the signs and then (12) uses polynomially many field and integer operations in q. Since $n = q^2$, the full rational matrix is obtained in polynomial bit complexity in n. □

The finite-field construction has constant distortion, but its rank is only $\sqrt{n}$, which gives the exponent $1/4$ through (5). To approach the square-root gap, we need a subspace with rank proportional to n while allowing a slowly growing distortion. This is the role of the construction in [4].

4 The near-square-root deterministic bound

Guruswami, Lee and Razborov [4, Theorem 1.1] give a deterministic polynomial-time construction of almost Euclidean subspaces. With their dimension-loss parameter fixed at $1/2$, the resulting spaces $E_n \subseteq \mathbb{R}^n$ satisfy, for sufficiently large n,

$$r_n = \dim E_n \geq \frac{n}{2}, \qquad \|x\|_1 \geq \frac{\sqrt{n}}{D_n}\|x\|_2 \quad (x \in E_n), \qquad D_n \leq (2\log\log n)^{O(\log\log n)}. \tag{19}$$

Their distortion is $\sqrt{n}\sup_{x\neq 0}\|x\|_2/\|x\|_1$, so no change of normalization is needed. All the subspace-construction input in (19), including its expander-code ingredients, is supplied by that theorem.

For rational output, we also need how the subspaces are specified. The local constraints in [4, Section 1.2.3, Theorem 3.3 and Section 4.3.2] are sign constraints, placed in selected coordinates and stacked. Common nonzero normalizing factors may be removed from individual constraint rows without changing their kernels. Thus the final space is $E_n = \ker B_n$, where B_n is an integer check matrix with entries in $\{0, 1, -1\}$ and polynomially many rows. Zero entries arise from placing local constraints in the ambient coordinates.

Fix a total deterministic version of this published constructor, including its finite-size branches and all choices within those branches. At a finite-size branch for which the construction's parameter requirements are not met, take B_n to have no rows, so $E_n = \mathbb{R}^n$. These choices are part of the fixed algorithm; they are not instructions to find an unspecified cutoff from the asymptotic inequality (19). The estimates in (19) are used only for sufficiently large n. If a finite exceptional output has zero-dimensional kernel, replace it by the same empty check matrix. This last condition is decided by exact row reduction and ensures a nonzero output projector for every n.

Select a maximal independent set of rows of B_n and put them into R_n. If its row rank is $s > 0$, set

$$P_n = I_n - R_n^{\mathsf{T}}(R_n R_n^{\mathsf{T}})^{-1} R_n; \tag{20}$$

if $s = 0$, set $P_n = I_n$. Since R_n has independent rows, its Gram matrix is positive definite. The second term in (20) is the orthogonal projector onto the row space of R_n, so P_n is the rational orthogonal projector onto E_n.

To bound the cost of exact arithmetic, write $G = R_n R_n^{\mathsf{T}}$. It is an $s \times s$ integer matrix with $s \leq n$ and entries of absolute value at most n. Hadamard's bound gives $O(n \log n)$ bits for its determinant and cofactors. The adjugate formula consequently bounds the numerator and denominator lengths of the entries of G^{-1}, and then of (20), by $O(n \log n)$. The same determinant bounds control intermediate sizes in fraction-free elimination. Row selection and the evaluation of (20) therefore require polynomially many bit operations. The adjugate formula is used here to bound sizes; the algorithm need not compute determinants by expansion.

Proof of Theorem 1. The preceding construction gives the stated rational projector and polynomial running time. For sufficiently large n, (19) and Lemma 2 give

$$\frac{\gamma_+^{\mathbb{F}}(P_n)}{\|P_n\|_{1,1}} \geq \frac{\sqrt{r_n}}{D_n^2} \geq \frac{\sqrt{n}}{\sqrt{2}D_n^2}. \tag{21}$$

Taking logarithms in the distortion bound shows that $\log D_n = O(\log\log n \log\log\log n)$. Enlarging the constant and the finite threshold absorbs the factor $\sqrt{2}$ and proves (3). The projector and constants are the same in both fields because (6) preserves the subspace constant. □

The estimate is a deterministic consequence of [4], with rationality supplied by the check-matrix representation. The algorithm produces matrices whose every factorization has a certified cost lower bound, without solving the minimization problem defining γ_+. Earlier explicit-section constructions can also be inserted into Lemma 2. In particular, [4, Section 1.1] describes an earlier construction of Indyk with essentially constant distortion and dimension $n/\exp[O((\log\log n)^2)]$, which gives a weaker subpolynomial loss by the same estimate. The specific rate in Theorem 1 uses the parameters in (19).

5 A specified infinite operator

The elementary family already provides enough growth to implement the block method of Balan and Jiang [1, Theorem 5.5]. We use it here because every dimension, weight and entry can be specified without the finite-size choices of the stronger construction. Let

$$q_j = 4^j, \qquad K_j = \mathbb{F}_{q_j}, \qquad H = \bigoplus_{j\geq 1} \ell_2(K_j^2), \qquad A = \bigoplus_{j\geq 1} q_j^{-3} P_{q_j}. \tag{22}$$

Choose each field and its coordinate order by the rule in Proposition 3, and concatenate the blocks in increasing j. This identifies H with $\ell_2(\mathbb{N})$. Writing Tr_j for the absolute trace on K_j, the within-block entries are

$$A_{(j,a,b),(j,c,d)} = q_j^{-5} \sum_{t\in K_j} (-1)^{\mathrm{Tr}_j((a+c)t+(b+d)t^3)}, \tag{23}$$

and entries between different blocks are zero. Thus every entry is rational and can be computed by a terminating deterministic procedure.

Theorem 4 (all countable factorizations have infinite cost). The operator A in (22) is positive and trace class, and

$$\sum_{u,v\geq 1} |A_{uv}| \leq 1. \tag{24}$$

If $f_k \in \ell_2(\mathbb{F})$ and

$$A = \sum_{k\geq 1} f_k f_k^* \tag{25}$$

in the weak operator topology, then $\sum_{k\geq 1} \|f_k\|_1^2 = \infty$, where $\|f_k\|_1 = \infty$ is allowed. This holds for both fields.

Proof. Each block is positive with operator norm q_j^{-3}, so (22) defines a bounded positive operator. Its trace and entrywise sum satisfy

$$\operatorname{tr} A = \sum_{j\ge 1} q_j^{-3} q_j = \sum_{j\ge 1} 16^{-j} = \frac{1}{15}, \qquad \|A\|_{1,1} \le \sum_{j\ge 1} q_j^{-3} q_j^{5/2} = \sum_{j\ge 1} 2^{-j} = 1. \tag{26}$$

Now let $Q_j : H \to \ell_2(K_j^2)$ be coordinate restriction to block j, and put $f_{k,j} = Q_j f_k$. Weak operator convergence in (25) gives the finite-dimensional identity

$$q_j^{-3} P_{q_j} = \sum_{k\ge 1} f_{k,j} f_{k,j}^*. \tag{27}$$

For a vector y in the kernel of P_{q_j}, its quadratic form in (27) is a sum of nonnegative numbers equal to zero. Hence every $f_{k,j}$ belongs to the range of P_{q_j} over the relevant field. Taking the finite-dimensional trace gives

$$\sum_{k\ge 1} \|f_{k,j}\|_2^2 = q_j^{-3} \operatorname{tr} P_{q_j} = q_j^{-2}. \tag{28}$$

The range inequality (17) therefore forces a cost of at least $1/3$ in each block:

$$\sum_{k\ge 1} \|f_{k,j}\|_1^2 \ge \frac{q_j^2}{3} \sum_{k\ge 1} \|f_{k,j}\|_2^2 = \frac{1}{3}. \tag{29}$$

For each k, the blocks partition the coordinates, so

$$\sum_{j\ge 1} \|f_{k,j}\|_1^2 \le \left(\sum_{j\ge 1} \|f_{k,j}\|_1 \right)^2 = \|f_k\|_1^2 \tag{30}$$

as an inequality of extended nonnegative quantities. Tonelli's theorem and (29) now imply

$$\sum_{k\ge 1} \|f_k\|_1^2 \ge \sum_{j\ge 1} \sum_{k\ge 1} \|f_{k,j}\|_1^2 \ge \sum_{j\ge 1} \frac{1}{3} = \infty. \tag{31}$$

This proves the assertion for every countable decomposition in (25). □

Weak operator convergence in the statement ensures that (27) follows by testing vectors supported on a single block. In particular, the theorem excludes every factorization with finite summed squared ℓ_1 cost: such a factorization would converge absolutely both in trace norm and in entrywise ℓ_1, since $\|ff^*\|_{S_1} = \|f\|_2^2 \le \|f\|_1^2$ and $\|ff^*\|_{1,1} = \|f\|_1^2$. The conclusion concerns all factors, not a failure of one chosen decomposition.

The elementary construction thus makes the infinite counterexample explicit, while the higher-rank subspaces in [4] supply the stronger finite-dimensional rate. Both conclusions follow from the same constraint on every positive rank-one factor. Removing the remaining subpolynomial loss from the deterministic bound would require a stronger quantitative input or an additional argument beyond the projector estimate used here.

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